HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2024)

Student No.: 10 Name: _____ Marks: / 40

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. Let $F(x) = (6x^2 + x + p)(qx^2 + rx - 10)$, where p, q and r are constants. The constant term of $F(x)$ is 40.

   (a) Write down the value of p. (1 mark)

   (b) When $F(x)$ is divided by $x + 1$, the remainder is -12. It is given that $x - 2$ is a factor of $F(x)$.

   (i) Find q and r.

   (ii) How many irrational roots does the equation $F(x) = 0$ have? Explain your answer. (7 marks)

(8 marks) (DSE-2024 PI Q14)

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2. The coordinates of the points Q and R are $(10, -1)$ and $(-4, -9)$ respectively.

(a) Let P be a moving point in the rectangular coordinate plane such that $PQ = PR$. Denote the locus of P by Γ.

(i) Describe the geometric relationship between Γ and QR.

(ii) Find the equation of Γ. (3 marks)

(b) Let C be the circle which passes through Q, R and the point $(4, 3)$.

(i) Find the equation of C.

(ii) The coordinates of the point U are $(10, 4)$. It is found that U lies outside C. UV and UW are the tangents to C at the points V and W respectively. Is the area of the circumcircle of $\triangle UVW$ greater than 100? Explain your answer. (5 marks)

(8 marks) (DSE-2024 PI Q17)

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3. (a) $PQRS$ is a thin quadrilateral metal sheet, where $PQ = 12\,\text{cm}$, $PS = 10\,\text{cm}$, $QR = 13\,\text{cm}$, $\angle QPS = 82^\circ$ and $\angle QRS = 65^\circ$. Find

(i) the length of QS,

(ii) $\angle RQS$. (4 marks)

(b) The metal sheet $PQRS$ described in (a) is now folded along QS (see Figure 2). It is given that the angle between the plane PQS and the plane QRS is 80°.

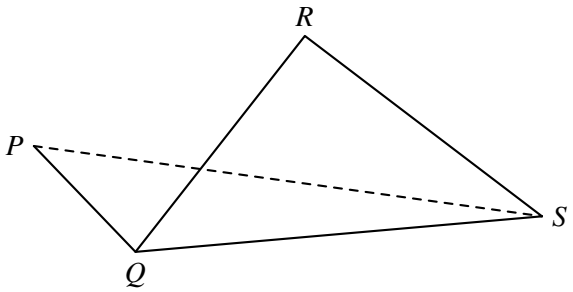


Figure 2

(i) Find the shortest distance from R to the plane PQS.

(ii) Let X be any point lying on the plane QRS. Someone claims that the distance between P and X exceeds $8\,\text{cm}$. Is the claim correct? Explain your answer. (4 marks)

(8 marks) (DSE-2024 PI Q18)

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4. In a school, 40% of the students are girls and $\beta\%$ of the girls are foreign students. It is given that 30% of the boys in the school are foreign students. In the school, the number of foreign students and the number of girls are equal. Find β.

A. 20

B. 45

C. 55

D. 80

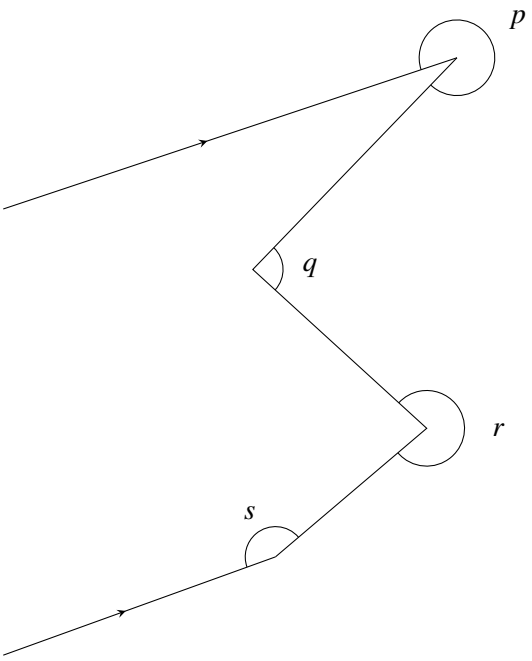
(2 marks) (DSE-2024 PII Q11)

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5. According to the figure, which of the following must be true?



A. $p+q-r=90^\circ$

B. $p-r+s=180^\circ$

C. $p+q-r+s=270^\circ$

D. $p+q+r-s=540^\circ$

(2 marks) (DSE-2024 PII Q19)

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6. If the sum of the interior angles of a regular polygon is 900°, which of the following is/are true?

I. The number of diagonals of the polygon is 7.

II. The number of folds of rotational symmetry of the polygon is 7.

III. The number of axes of reflectional symmetry of the polygon is 7.

A. I only

B. II only

C. I and III only

D. II and III only

(2 marks) (DSE-2024 PII Q20)

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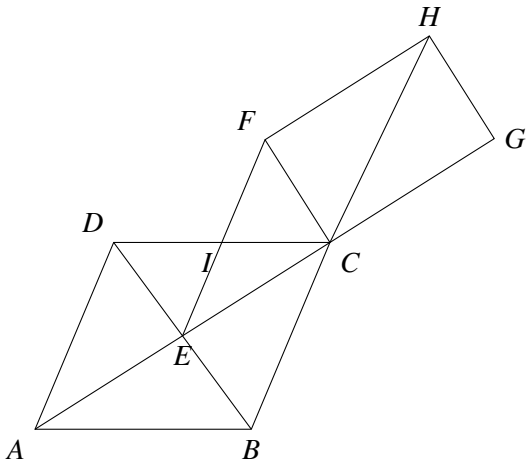
Answers written in the margins will not be marked.

7. In the figure, $ABCD$ is a rhombus. Denote the point of intersection of AC and BD by E. Let F be a point such that $BH//EF$ and $CFHG$ is a rectangle, where G and H are points lying on AC produced and BC produced respectively. Denote the point of intersection of CD and EF by I. Which of the following must be true?

I. $CI = FI$

II. $\angle ABE = \angle GCH$

III. $\triangle ADE \cong \triangle HCF$



A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(2 marks) (DSE-2024 PII Q21)

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8. Two numbers are randomly drawn at the same time from six cards numbered 1, 2, 3, 4, 5 and 6 respectively. Find the probability that the product of the numbers drawn is not less than 12.

A. $\frac{1}{3}$

B. $\frac{2}{3}$

C. $\frac{7}{15}$

D. $\frac{8}{15}$

(2 marks) (DSE-2024 PII Q28)

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9. $AF000000000BC_{16} =$

A. $175 \times 16^{11} + 188$.

B. $192 \times 16^{11} + 205$.

C. $175 \times 16^{12} + 188$.

D. $192 \times 16^{12} + 205$.

(2 marks) (DSE-2024 PII Q32)

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10. Consider the following system of inequalities:

$$\begin{cases} x - 2y \leq 1 \\ x + 4y \leq 13 \\ 2x - y \geq -1 \end{cases}$$

Let R be the region which represents the solution of the above system of inequalities. Find the constant c such that the least value of $5x - 2y + c$ is 22, where (x, y) is a point lying in R.

A. 1

B. 23

C. 25

D. 29

(2 marks) (DSE-2024 PII Q37)

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11. Let G, H, I and J be the centroid, the orthocentre, the in-centre and the circumcentre of $\triangle PQR$ respectively. If $\angle PQR = \angle PRQ = 22°$, which of the following are true?

I. G lies inside $\triangle PQR$.

II. H lies outside $\triangle PQR$.

III. I, J and Q are collinear.

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(2 marks) (DSE-2024 PII Q41)

I can finish _____ questions

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